

Prasanth Muntha

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SUMMARY

Graduate student in Artificial Intelligence with experience applying machine learning and computer vision to healthcare and clinical data. Interested in supporting research, data analysis, and computational modeling in academic and biomedical settings.

EDUCATION

Yeshiva University

Master of Science in Artificial Intelligence

New York, NY

Exp. Dec 2026

JNTU, India

Bachelor of Science in Computer Science

Hyderabad, India

May 2023

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Bash

Machine Learning & AI: Machine Learning, Deep Learning, NLP, Computer Vision, BERT, LLMs, RAG, LangChain

ML & CV Frameworks: PyVista, VTK, MoveNet, ONNX Runtime, OpenCV, YOLOv8, Scikit-learn, Pandas

Data Engineering: Apache Airflow, AWS Glue, Athena, S3, ETL Pipelines

Backend & APIs: FastAPI, Flask, REST APIs, GraphQL, Streamlit

Cloud & DevOps: AWS, Docker, Git

WORK EXPERIENCE

Aloha Technologies

Software Developer

India

2022 – 2025

- **3D Scoliosis Detection System:** Built a production-grade 3D AI inference pipeline integrating geometric computer vision with deep learning to automate scoliosis diagnosis from raw 3D body scans.
- Designed a mesh rectification algorithm using Oriented Bounding Boxes and K-Means clustering to correct scan misalignment and eliminate manual preprocessing.
- Implemented semantic orientation detection with ONNX Runtime + MoveNet, achieving **99% accuracy** in anatomical view classification.
- Developed ROI segmentation using statistical regression to suppress surface noise and improve Cobb Angle measurement accuracy.
- **AWS Airflow Data Migration:** Migrated legacy Pentaho workflows to Amazon MWAA, refactoring monolithic ETL jobs into modular Python-based DAGs.
- Improved pipeline reliability and fault tolerance, reducing runtimes by **40%**.
- **High-Throughput Data Ingestion:** Built ingestion pipelines processing millions of XML and CSV records daily with **sub-200ms latency**.
- Developed a real-time GraphQL monitoring dashboard, reducing engineering support overhead.

PROJECTS

GitHub Codebase Decoder (Ongoing)

Python, LLMs, Static Analysis, RAG, AWS

- Building a system to automatically analyze, summarize, and explain large codebases using AST parsing, embeddings, and retrieval-augmented generation.
- Designed to help researchers and students understand unfamiliar repositories and software architecture quickly.

Local-First Agentic RAG & Data Analyst

Python, LangChain, Llama 3.2, FastAPI, ChromaDB

- Built a privacy-first local AI platform using Ollama and Llama 3.2 with air-gapped inference and zero data leakage.
- Implemented agentic query routing across vector search, web search, and Python-based data analysis.
- Developed a Data Analyst agent that infers schemas and generates interactive visualizations using Pandas and Streamlit.

Agentic Fake News Detection System

Python, BERT, Llama-3-8B

- Designed a hybrid verification pipeline combining BERT classification with LLM-based semantic fact-checking.
- Improved classification reliability by leveraging LLM reasoning to explain prediction outcomes.